Homework 1

Due Date: March 30 (Mon)

1. Convert the following problems to standard form:

(a)

$$\begin{aligned} \min \quad & 2x + 5y - 3z \\ \text{s.t.} \quad & -3 \leq 2x + y \leq 5 \\ & 4x - z \geq 2 \\ & x \geq 0, y \geq 0. \end{aligned}$$

(b)

$$\begin{aligned} \min \quad & 4x - y + 2z \\ \text{s.t.} \quad & x + 3y - 2z = 7 \\ & 2x - y + z \leq 10 \\ & x \geq 2, y \leq 1, z \geq 0. \end{aligned}$$

Solution to P1.

(a)

$$\begin{aligned} \max \quad & -2x - 5y + 3z_1 - 3z_2 \\ \text{s.t.} \quad & 2x + y \leq 5 \\ & -(2x + y) \leq 3 \\ & -4x + z_1 - z_2 \leq -2 \\ & x, y, z_1, z_2 \geq 0 \end{aligned}$$

(b)

$$\begin{aligned} \max \quad & -4\hat{x} - \hat{y} - 2z \\ \text{s.t.} \quad & \hat{x} - 3\hat{y} - 2z \leq 2 \\ & -\hat{x} + 3\hat{y} + 2z \leq -2 \\ & 2\hat{x} + \hat{y} + z \leq 7 \\ & \hat{x}, \hat{y}, z \geq 0 \end{aligned}$$

2. Convert the following problem to a linear program:

(a)

$$\begin{aligned} \min \quad & 2|x-1| + |y+2| + 3|z| \\ \text{s.t.} \quad & x - 2y \leq 4 \\ & x + y + z = 5 \end{aligned}$$

(b)

$$\begin{aligned} \min_{x} \quad & \frac{c^T x + d}{e^T x + f} \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

Assume that $e^T x + f > 0$ for any x that satisfies $Ax \leq b$.

Solution to P2.

(a) Solution (1):

$$\begin{aligned} \max \quad & -2x_1 - 2x_2 - y_1 - y_2 - 3z_1 - 3z_2 \\ \text{s.t.} \quad & x_1 - x_2 - 2y_1 + 2y_2 \leq -1 \\ & x_1 - x_2 + y_1 - y_2 + z_1 - z_2 \leq 6 \\ & -x_1 + x_2 - y_1 + y_2 - z_1 + z_2 \leq -6 \\ & x_1, x_2, y_1, y_2, z_1, z_2 \geq 0 \end{aligned}$$

Solution (2):

$$\begin{aligned} \min \quad & 2u_1 + u_2 + 3u_3 \\ \text{s.t.} \quad & u_1 - x \geq -1 \\ & u_1 + x \geq 1 \\ & u_2 - y \geq 2 \\ & u_2 + y \geq -2 \\ & u_3 - z \geq 0 \\ & u_3 + z \geq 0 \\ & x - 2y \leq 4 \\ & x + y + z = 5 \\ & u_1, u_2, u_3 \geq 0 \\ & x, y, z \in \mathbb{R} \end{aligned}$$

(b) Define $\boldsymbol{y} = \frac{\boldsymbol{x}}{\boldsymbol{e^T x + f}}, \quad z = \frac{1}{\boldsymbol{e^T x + f}}$, then

$$\begin{aligned} \min_{\boldsymbol{y}, \boldsymbol{z}} \quad & \boldsymbol{c}^T \boldsymbol{y} + dz \\ \text{s.t.} \quad & A\boldsymbol{y} - bz \leq 0 \\ & \boldsymbol{e}^T \boldsymbol{y} + fz = 1 \\ & z \geq 0 \end{aligned}$$

3. Consider the following model:

$$
\begin{aligned}
\min \quad & Z = 40x + 50y \\
\text{s.t.} \quad & 2x + 3y \geq 30 \\
& x + y \geq 12 \\
& 2x + y \geq 20 \\
& x \geq 0, y \geq 0.
\end{aligned}
$$

(a) Use the graphical method to solve this model.

(b) How does the optimal solution change if the objective function is changed to $Z = 40x + 70y$?

(c) How does the optimal solution change if the third functional constraint is changed to $2x + y \geq 15$?

Solution to P3.

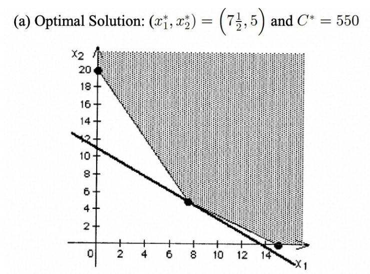

Figure 1: Solution to the Q1.

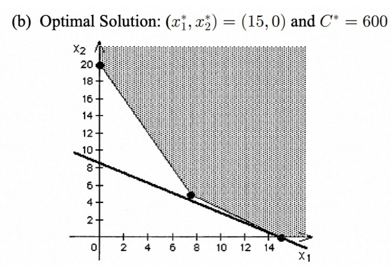

Figure 2: Solution to the Q2.

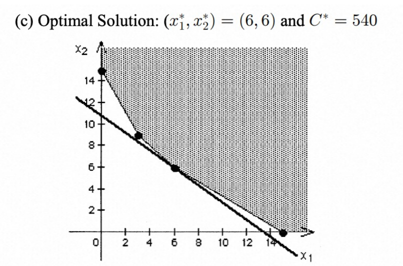

Figure 3: Solution to the Q3.

4. Fred Jonasson manages a family-owned farm. To supplement several food products grown on the farm, Fred also raises pigs for market. He now wishes to determine the quantities of the available types of feed (corn, tankage, and alfalfa) that should be given to each pig. Since pigs will eat any mix of these feed types, the objective is to determine which mix will meet certain nutritional requirements at a minimum cost. The number of units of each type of basic nutritional ingredient contained within a kilogram of each feed type is given in the following table, along with the daily nutritional requirements and feed costs:

Nutritional Ingredient	Corn (kg)	Tankage (kg)	Alfalfa (kg)	Minimum Daily Requirement
Carbohydrates	90	20	40	200
Protein	30	80	60	180
Vitamins	10	20	60	150
Cost (Φ)	84	72	60	-

Formulate a linear programming model for this problem.

Solution to P4.

$$
\begin{aligned}
\min \quad & C = 84C + 72T + 60A \\
\text{s.t.} \quad & 90C + 20T + 40A \geq 200 \\
& 30C + 80T + 60A \geq 180 \\
& 10C + 20T + 60A \geq 150 \\
\text{and} \quad & C, T, A \geq 0
\end{aligned}
$$

5. Joyce and Marvin run a day care for preschoolers. They are trying to decide what to feed the children for lunches. They would like to keep their costs down, but also need to meet the nutritional requirements of the children. They have already decided to go with peanut butter and jelly sandwiches, and some combination of graham crackers, milk, and orange juice. The nutritional content of each food choice and its cost are given in the table below.

Food	**Calories from Fat**	**Total Calories**	**Vitamin C (mg)**	**Protein (g)**	**Cost**
Bread (1 slice)	10	70	0	3	5
Peanut butter (1tbsp)	75	100	0	4	4
Straberry jelly (1 tbsp)	0	50	3	0	7
Graham cracker (1 cracker)	20	60	0	1	8
Milk (1 cup)	70	150	2	8	15
Juice (1 cup)	0	100	120	1	35

The nutritional requirements are as follows. Each child should receive between 400 and 600 calories. No more than 30 percent of the total calories should come from fat. Each child should consume at least 60 milligrams (mg) of vitamin C and 12 grams (g) of protein. Furthermore, for practical reasons, each child needs exactly 2 slices of bread (to make the sandwich), at least twice as much peanut butter as jelly, and at least 1 cup of liquid (milk and/or juice). Joyce and Marvin would like to select the food choices for each child which minimize cost while meeting the above requirements.

Formulate a linear programming model for this problem.

Solution to P5.

Let B = slices of bread, P = tablespoons of peanut butter, S = tablespoons of strawberry jelly, G = graham crackers, M = cups of milk, and J = cups of juice.

$$
\begin{array}{lc}
\text{min} & C = 5B + 4P + 7S + 8G + 15M + 35J \\
\text{s.t.} & 70B + 100P + 50S + 60G + 150M + 100J \geq 400 \\
 & 70B + 100P + 50S + 60G + 150M + 100J \leq 600 \\
 & 10B + 75P + 20G + 70M \leq 0.3(70B + 100P + 50S + 60G + 150M + 100J) \\
 & 3S + 2M + 120J \geq 60 \\
 & 3B + 4P + G + 8M + J \geq 12 \\
 & B = 2 \\
 & P \geq 2S \\
 & M + J \geq 1 \\
\text{and} & B, P, S, G, M, J \geq 0
\end{array}
$$

6. Larry Edison is the director of the Computer Center for Buckly College. He now needs to schedule the staffing of the center. It is open from 8 a.m. until midnight. Larry has monitored the usage of the center at various times of the day, and determined that the following number of computer consultants are required

Time of Day	Minumum Number of Consultants Required to be On Duty
8 a.m. - noon	4
noon - 4 p.m.	8
4 p.m. - 8 p.m.	10
8 p.m. - midnight	6

Two types of computer consultants can be hired: full-time and part-time. The full-time consultants work for 8 consecutive hours in any of the following shifts: morning (8 a.m.–4 p.m.), afternoon (noon–8 p.m.), and evening (4 p.m.–midnight). Full-time consultants are paid \$40 per hour.

Part-time consultants can be hired to work any of the four shifts listed in the above table. Part-time consultants are paid \$30 per hour.

An additional requirement is that during every time period, there must be at least 2 full-time consultants on duty for every parttime consultant on duty.

Larry would like to determine how many full-time and how many part-time workers should work each shift to meet the above requirements at the minimum possible cost.

Please formulate a linear programming model for this problem.

Solution to P6.

Let

f_1 = number of full-time consultants working the morning shift (8 a.m.-4 p.m.),
f_2 = number of full-time consultants working the afternoon shift (Noon-8 p.m.),
f_3 = number of full-time consultants working the evening shift (4 p.m.-midnight),
p_1 = number of part-time consultants working the first shift (8 a.m.-noon),
p_2 = number of part-time consultants working the second shift (Noon-4 p.m.),
p_3 = number of part-time consultants working the third shift (4 p.m. -8 p.m.),
p_4 = number of part-time consultants working the fourth shift (8 p.m.-midnight).

$$
\begin{array}{ll}
\min & C = (40 \times 8)\,(f_1 + f_2 + f_3) + (30 \times 4)\,(p_1 + p_2 + p_3 + p_4) \\
\text{s.t.} & f_1 + p_1 \geq 4 \\
& f_1 + f_2 + p_2 \geq 8 \\
& f_2 + f_3 + p_3 \geq 10 \\
& f_3 + p_4 \geq 6 \\
& f_1 \geq 2p_1 \\
& f_1 + f_2 \geq 2p_2 \\
& f_2 + f_3 \geq 2p_3 \\
& f_3 \geq 2p_4 \\
& f_1, f_2, f_3, p_1, p_2, p_3, p_4 \geq 0
\end{array}
$$